YEAR 7 · UNIT 2 · EXTRA HOMEWORK

One Letter at a Time

Expressions, substituting · and where volume comes from

Name: ______________________ **Class:** ______________________

Date given: __________________ **Date due:** Monday 14 September 2026

Why this sheet exists

There is almost **no arithmetic** in Unit 2. Every number in it is small. What decides the answer is **one letter** — what it stands for, and where you put it.

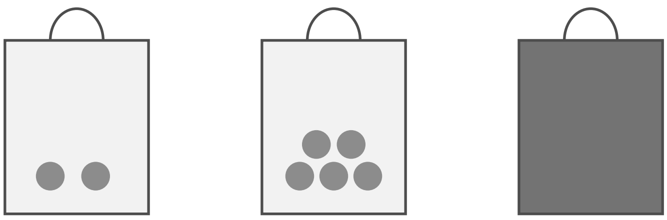
So this sheet does one letter at a time, and it shows you the **middle line** of every calculation with the numbers left out. Filling in those boxes is not extra working. It **is** the method.

No calculator. Write in every box.

Part 1 · What a letter is

E1 The bag you cannot see into

Write the number of counters in each bag. For the last one you cannot count them, so **choose your own letter** instead.



(a) (b) (c)

(a) ☐ counters (b) ☐ counters (c) ☐ counters

Why (c) needs a letter

You cannot see inside bag (c), so nobody knows the number. That does not mean there is no number — there is one, and you just cannot count it. So we give it a **letter** instead, and carry on.

E2 Put three more in

Three counters are put into each bag. Fill in the boxes.

(a) There are **6** counters in the bag.

Put in three more, so now there are $6 + 3 = \square$ counters.

(b) There are **10** counters in the bag.

Put in three more, so now there are $\square + 3 = \square$ counters.

(c) There are n counters in the bag.

Put in three more, so now there are $\square + 3$ counters.

(d) There are n counters in the bag.

Take four out, so now there are $\square - \square$ counters.

Part (c) is the one that matters

You **cannot** work out $n + 3$. There is nothing to work out — you do not know what n is. So you **leave it as it is**. $n + 3$ is the **finished answer**, and this is called an **expression**.

E3 Say it the short way

The three rules

$4 \times n$ is written **4n** — the number goes **in front**.
$a \times b$ is written **ab**.
$w \div 2$ is written **w over 2**, as a fraction.

The first one is done for you.

(a) $4 \times n$ = **4n**

(b) $7 \times k$ = __________

(c) $m \times p$ = __________

(d) $3 \times a \times b$ = __________

(e) $w \div 2$ = __________

(f) $t \div 5$ = __________

E4 One beaker, four things to do to it

Mr Bowen has a beaker holding b **cm^3** of water. Nobody has measured it, so we call it b.

Which operation?

pours in more	add
pours some out	subtract
three beakers the same	multiply
half of it	divide by 2

(a) He pours in 40 cm^3 more. How much is in the beaker now? ____________

(b) He pours out 25 cm^3. How much is left? ____________

(c) He has **three** beakers, all the same. How much altogether? ____________

(d) He pours away **half** of the first beaker. How much is left? ____________

E5 Match the description to the expression

Every description uses the **same** two numbers, 3 and 8. Only the **order** changes. Write the correct **roman numeral** in each box. One expression is left over. The first one is done for you: **a** goes with **i**.

	Description			Expression
a	Multiply x by 3, then add 8.	i	i	$3x + 8$
b	Multiply x by 8, then add 3.		ii	$8x - 3$
c	Multiply x by 3, then subtract 8.		iii	$3 - 8x$
d	Multiply x by 3, then subtract the result **from** 8.		iv	$8x + 3$
e	Multiply x by 8, then subtract 3.		v	$3x - 8$
f	Multiply x by 8, then subtract the result **from** 3.		vi	$8 - 3x$
			vii	$8x - 8$

The one word to watch

subtract 8 means take 8 away from what you have.
subtract from 8 means **start** at 8 and take away what you have.
Those two give different answers. The word is from.

Part 2 · Putting a number back in

E6 Expression or formula?

One difference, and it is the equals sign

An **expression** has letters and **no** equals sign: $7k$, $m + 9$.
A **formula** has letters **and** an equals sign: $A = lw$, $y = 3x$.

Write **E** for expression or **F** for formula. The first two are done for you.

(a) $7k$	**E**	(e) $5p - 2$	______
(b) $y = 3x$	**F**	(f) $V = 4t$	______
(c) $m + 9$	______	(g) $2a + 3b$	______
(d) $A = lw$	______	(h) $P = 2l + 2w$	______

E7 The same expression, four times

Only the number changes. Work down each list. The first one in each is done for you.

(a) The value of $m + 7$

when $m = 1$: $1 + 7 =$ **8**

when $m = 2$: $\square + 7 = \square$

when $m = 5$: $\square + 7 = \square$

when $m = 9$: $\square + 7 = \square$

(b) The value of $4m$

when $m = 1$: $4 \times 1 =$ **4**

when $m = 2$: $4 \times \square = \square$

when $m = 5$: $4 \times \square = \square$

when $m = 9$: $4 \times \square = \square$

Look at column (b) again

$4m$ means $4 \times m$. The times sign was hidden, not deleted.
So $4m$ with $m = 2$ is $4 \times 2 =$ **8**. **It is never 42.**

E8 The middle line, with the numbers left out

Every middle line below is already written for you. All you have to do is fill in the boxes, left to right.

(a) $5n$ when $n = 7$: $5n = 5 \times \square = \square$

(b) $n + 12$ when $n = 7$: $n + 12 = \square + 12 = \square$

(c) $3x + 4$ when $x = 6$: $3 \times \square + 4 = \square + 4 = \square$

(d) $20 - 2n$ when $n = 8$: $20 - 2 \times \square = 20 - \square = \square$

(e) $\dfrac{k}{4}$ when $k = 36$: $\square \div 4 = \square$

(f) $2a + 3b$ when $a = 5, b = 2$: $2 \times \square + 3 \times \square = \square + \square = \square$

Parts (c) and (d)

In both of them the **times** happens before the + or the –. That is why the middle line has three steps and not two — the multiplying is a step of its own, and it comes first.

E9 True or false — and if it is false, put it right

The first one is done for you.

Statement	True or false?	If false, the correct value is
(a) The value of $3m$ when $m = 4$ is 34.	False	$3 \times 4 = 12$
(b) The value of $2p$ when $p = 9$ is 18.		
(c) The value of $n + 5$ when $n = 11$ is 16.		
(d) The value of $6k$ when $k = 3$ is 63.		
(e) The value of $\frac{w}{3}$ when $w = 21$ is 7.		
(f) The value of $4 + 2t$ when $t = 5$ is 30.		

If a statement is false, **write the middle line** in the last column, the way you did in **E8**. That is where the mistake shows.

E10 Two formulae, one rectangle

The two rules for a rectangle

Area: $A = lw$ — that is $l \times w$.
Perimeter: $P = 2l + 2w$ — the distance all the way round the edge.

Fill in the table. The first row is done for you.

l	w	$A = lw$	$P = 2l + 2w$
5	3	$5 \times 3 = 15$	$10 + 6 = 16$
8	2		
10	6		
7	7		

Part 3 · Where volume comes from

In Science you measured water in a cylinder, in cm^3. Here is where that little **3** comes from.

Count the lengths you multiplied

One length: cm

A length **times** a length: cm $\times$ cm = cm^2

A length **times** a length **times** a length: cm $\times$ cm $\times$ cm = cm^3

The small number is just **how many lengths you multiplied**. Nothing else.

And the unit that joins the two subjects: 1 cm^3 = 1 ml, 1000 cm^3 = 1 litre.

E11 Count the letters, then write the unit

Every letter stands for a length in centimetres. The first **two** rows are done for you.

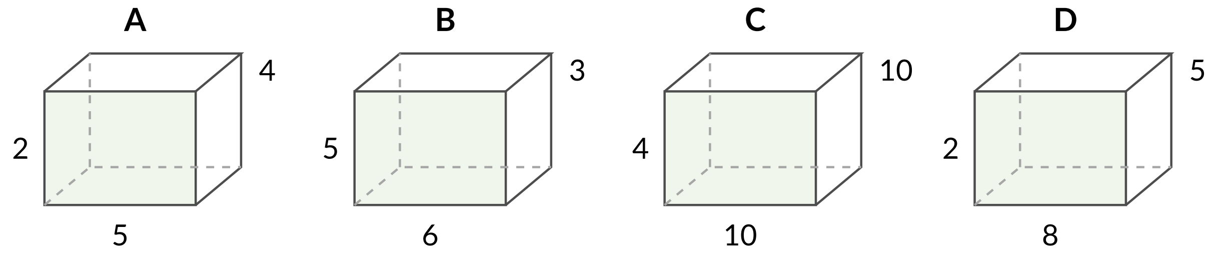
Expression	How many lengths multiplied?	Unit
h	one	cm
lw	two	cm^2
lwh		
$3ab$		
$s \times s \times s$		

E12 Find the volume of each box

One formula

$V = lwh$ means length $\times$ width $\times$ height. Three measurements, multiplied.

The first one is done for you.



A $V = 5 \times 4 \times 2 = \mathbf{40\ cm^3}$

B $V = \square \times \square \times \square = \square\ \text{cm}^3$

C $V = \square \times \square \times \square = \square\ \text{cm}^3$

D $V = \square \times \square \times \square = \square\ \text{cm}^3$

E13 The tank, one step at a time

A tank has a rectangular base measuring **20 cm** by **10 cm**. Mr Bowen pours in water to a depth of h centimetres. Nobody has measured the depth, so we call it h.

(a) First, the **base**. Its area is $20 \times 10 = \square\ \text{cm}^2$

(b) Now the water. Its volume is the base area **times** the depth:

$V = \square \times h$

(c) Write that the short way, with the number in front of the letter:

$V =$ ______________

(d) Now use it. When $h = 3$: $V = \square \times 3 = \square\ \text{cm}^3$

(e) And when $h = 7$: $V = \square \times \square = \square\ \text{cm}^3$

E14 Cubic centimetres, millilitres and litres

Science measures water in cm^3 and a bottle is labelled in ml. They are the same thing.

Two facts, and that is all

$1\ \text{cm}^3 = 1$ ml — exactly the same amount, two names.
$1000\ \text{cm}^3 = 1$ litre

(a) $250\ \text{cm}^3 =$ _________ ml

(b) $2000\ \text{cm}^3 =$ _________ litres

(c) $1500\ \text{cm}^3 =$ _________ litres

(d) 3 litres = _________ cm^3

(e) A water bottle holds 500 ml. A measuring cylinder in the science lab is marked in cm^3. How many cm^3 of water does the bottle hold? ________________

(f) The tank in **E13** holds $200h$ cm^3 of water. Mr Bowen fills it to a depth of $h = 25$ cm. How many **litres** is that?

E15 Say what changed

Mr Bowen pours the **same** 200 cm^3 of water from a tall narrow cylinder into a wide flat dish. Nothing is added and nothing is spilled.

Tick one box in each row.

	Changes	**Stays the same**
How **long** the water is		
How **wide** the water is		
How **deep** the water is		
The **volume** of the water		

Now finish the sentence. Use each word **once**: **shape** **volume**

A liquid keeps the same ________________ but takes the ________________ of its container.

Answer Key — Teacher Copy

E1 The bag you cannot see into

(a) **2** (b) **5** (c) **any letter**, with n, b and x all equally right.

The answer to (c) is a letter, and *which* letter does not matter — say that out loud, because a student who thinks there is a correct letter to guess has misunderstood the whole idea. What does matter is that they wrote something. A blank in (c) means the student believes a question with no number has no answer, and that belief will cost them every mark in Unit 2.

E2 Put three more in

(a) 6 + 3 = **9** (b) **10** + 3 = **13** (c) **n** + 3 (d) **n** − **4**

The ladder is the point: (a) is arithmetic, (b) is the same arithmetic with the first number handed over, and (c) is the same sentence again with a letter in it. Nothing changed except that we stopped being told the number.

Expect (c) to be answered **n3** or left blank. Neither is a maths error. The first is notation and is fixed in E3; the second is the belief that an answer must be a single number, and it is fixed by saying *you are allowed to stop* as often as it takes.

E3 Say it the short way

(b) $7k$ (c) mp (d) $3ab$ (e) $\frac{w}{2}$ (f) $\frac{t}{5}$

Watch for $k7$ in (b). The number goes in front, always, and it is worth correcting on sight rather than in writing.

Accept $w \div 2$ for (e) and (f) today, but write the fraction beside it — the workbook and the exam both use the fraction, and a student who has only ever seen ÷ will not recognise it.

E4 One beaker, four things to do to it

(a) $b + 40$ (b) $b - 25$ (c) $3b$ (d) $\frac{b}{2}$

All four answers are unfinished, and all four are correct. This is the same question the workbook asks about a box of toys, moved into the beaker the class handled in Science, so the letter has something real behind it.

If a student invents a number for b and answers everything in numbers, they have done four correct calculations and answered none of the questions. Do not mark it wrong — ask them what b is, wait, and let them arrive at “we do not know”.

E5 Match the description to the expression

a–**i** b–**iv** c–**v** d–**vi** e–**ii** f–**iii**. Left over: **vii** ($8x - 8$).

This is the best question on the sheet. Every description uses 3 and 8 and nothing else, so there is no arithmetic anywhere in it — the only thing being tested is reading, which is exactly the barrier in this class.

The pairs to mark together are **c and d** ($3x - 8$ against $8 - 3x$) and **e and f** ($8x - 3$ against $3 - 8x$). One word, *from*, is the entire difference. A student who gets c and e right and d and f wrong has read every word except that one, and that is a very specific, very fixable diagnosis.

The left-over expression matters too: a student who forces a match for vii has assumed every box must be used, which is the same instinct as inventing a number for b.

E6 Expression or formula?

(c) E (d) F (e) E (f) F (g) E (h) F

Six items, one rule, thirty seconds. It is here to be got right — after E5 the class needs a question it can finish.

The only wobble is (h), $P = 2l + 2w$, because it is longer than the others. Length is not the test; the equals sign is.

E7 The same expression, four times

(a) $m = 2 \to 9$, $m = 5 \to 12$, $m = 9 \to 16$.
(b) $m = 2 \to 8$, $m = 5 \to 20$, $m = 9 \to 36$.

Column (b) is the whole reason the two lists are printed side by side. In (a) the letter is replaced and the sentence reads the same way; in (b) the letter is replaced and a hidden times sign has to come back.

The errors to expect are **42**, **45** and **49** in column (b) — the digits pushed together. A student making that error is not confused about multiplication; they are reading $4m$ as a two-digit number. Write $4 \times m$ over the top of it on their page.

E8 The middle line, with the numbers left out

(a) $5 \times 7 =$ **35**

(b) $7 + 12 =$ **19**

(c) $3 \times 6 + 4 = 18 + 4 =$ **22**

(d) $20 - 2 \times 8 = 20 - 16 =$ **4**

(e) $36 \div 4 =$ **9**

(f) $2 \times 5 + 3 \times 2 = 10 + 6 =$ **16**

This is the sheet in miniature. The middle line is printed with the numbers missing, so it cannot be skipped — filling the boxes *is* the method. A student who can fill these in and still writes 34 for $3n$ on the main packet has not transferred the habit, and the fix is to make them draw the boxes themselves for a week.

(d) is the one that goes wrong. The multiplication is written second and happens first; a student working strictly left to right gets $18 \times 8 = 144$. The printed skeleton makes that almost impossible, which is the point — and is also why the same question without the skeleton belongs on the next sheet, not this one.

E9 True or false

(b) **True** (c) **True** (d) **False**; $6 \times 3 =$ **18** (e) **True** (f) **False**; $4 + 2 \times 5 = 4 + 10 =$ **14**

Three of the five are true, deliberately. A sheet where everything is wrong teaches students to answer *false* without reading.

(d) is the digits-pushed-together error again, and it should now be spotted instantly after E7.

(f) is the order-of-operations error, and it is the harder one: 30 comes from doing $4 + 2 = 6$ and then 6×5. Both numbers in that wrong answer are real numbers from the question, which is what makes it convincing. Ask the student to write the middle line from E8 and it collapses on its own.

E10 Two formulae, one rectangle

l	w	$A = lw$	$P = 2l + 2w$
5	3	$5 \times 3 = 15$	$10 + 6 = 16$
8	2	$8 \times 2 = 16$	$16 + 4 = 20$
10	6	$10 \times 6 = 60$	$20 + 12 = 32$
7	7	$7 \times 7 = 49$	$14 + 14 = 28$

The 8-by-2 row is worth pointing at: the area and the perimeter are 16 and 20, two numbers of similar size, and a student who has muddled the two formulae will not notice from the answer. The 10-by-6 row separates them properly.
The 7-by-7 row is a square. Nobody has to be told; it is there so that somebody notices, and if a student says "that one is a square" the lesson has landed.

E11 Count the letters, then write the unit

lwh	three	cm^3
$3ab$	**two** — the 3 is not a length	cm^2
$s \times s \times s$	three	cm^3

$3ab$ is the row that teaches something. There are three symbols and only *two* of them are lengths — 3 is just a count, so it does not carry a centimetre. Three rectangles of area ab is still an area.
Expect cm^3 for that row from most of the class, and treat it as a discussion rather than a cross: the rule they extracted, "count the things being multiplied", is a reasonable rule and it is nearly right. The correction is one word: count the *lengths*.

E12 Find the volume of each box

A = $5 \times 4 \times 2$ = **40 cm^3** **B** = $6 \times 3 \times 5$ = **90 cm^3**
C = $10 \times 10 \times 4$ = **400 cm^3** **D** = $8 \times 5 \times 2$ = **80 cm^3**

All four boxes are drawn the same size on purpose. The picture carries no information at all, so the numbers have to be read — and a student who answers "C is the biggest because it looks biggest" has been caught by a trap that was set for exactly that.
Box C is 400, not 40: 10×10 is where the zeros get dropped.
If anyone answers **11** for A, they added the three lengths. That is C5.3 on the main packet, and it is the reason the unit says 3.

E13 The tank, one step at a time

(a) 20×10 = **200 cm^2** (b) $V = $ **200** $\times h$ (c) $V = $ **200h**
(d) $V = 200 \times 3$ = **600 cm^3** (e) $V = 200 \times 7$ = **1400 cm^3**

The base area is worked out *first*, and as a number, before the letter ever appears. That is the whole design: by the time h arrives there is only one multiplication left to write, and $200 \times h$ is not frightening.
(c) is the only step with no arithmetic in it and it is the one to check — writing $200 \times h$ as $200h$ is E3 arriving in a new place, and a student who writes $h200$ has the maths right and the notation wrong.
(d) is where $200h$ with $h = 3$ could come out as **2003**. The printed skeleton prevents it here. On the main packet, in C2, it does not.

E14 Cubic centimetres, millilitres and litres

(a) **250** ml (b) **2** litres (c) **1.5** litres (d) **3000 cm^3**
(e) **500 cm^3** (f) $200 \times 25 = 5000$ cm^3 = **5** litres.

(a) is meant to look like a trick and is not one: the number does not change, because a cubic centimetre and a millilitre are the same amount of space with two different names. Some students will not believe it. Show them a 1 cm^3 cube and a 1 ml mark on a syringe.
(c) is the only answer on the sheet that is not a whole number.
(f) uses the formula built in E13, so it is worth the space. A student who gets 5000 and stops has done the maths and not read the word *litres* — which is, as always, the actual difficulty.

E15 Say what changed

How long — **changes** How wide — **changes** How deep — **changes** The volume — **stays the same**

A liquid keeps the same **volume** but takes the **shape** of its container.

This is the only question on the sheet with no arithmetic in it, and it is the one to talk about. It is the science property from 2.1a and an algebraic fact at the same time: three measurements all change, and their product does not. Neither subject could have said that on its own.
The row that decides it is the fourth. A student who ticks *changes* for the volume has not accepted the sentence they wrote underneath, and pouring 200 cm^3 between a cylinder and a dish at the front of the room settles it in twenty seconds.
Expect *stays the same* for "how long" from a few, on the grounds that it is the same water. Ask them to hold their hands apart to show the water in the cylinder, then in the dish.